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First in India

SPATIAL BIOLOGY SOLUTIONS @theraCUES™

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Resolve Heterogeneity of Tissue Spatially Through GeoMx™ DSP

IMAGINE . INNOVATE . IMPACT

Image courtesy ; NanoString Technologies

GeoMx™ DSP Enables Spatial High-Plex Protein & RNA Profiling

Spatial NGS

The GeoMx™ DSP from, nanoString, is a platform that enables high-plex spatial and molecular analysis of RNA and protein in tissue specimens on slides.

Tissue slides are stained with large panels of pre-mixed unique Biological Probes (RNAinsitu or Antibodies crosslinked with UV-cleavable barcode oligos) which can bind to mRNA or proteins specifically; and fluorescently labeled Morphology Reagents (bind to specific targets on the tissue) to determine the cell type/tissue morphology).

Using the morphology markers, ROIs are selected and are illuminated with UV light. The UV cleaved barcodes/oligos from these ROIs are collected and counted in an nCounter platform (NanoString) or sequenced using a Next Generation Sequencer (Illumina).

The resulting counts constitute Protein/RNA expression of key targets expressed in different ROIs and elucidate the biology of corresponding ROIs.

Presently, GeoMx DSP can profile upto 22,000+ mRNA targets and 140+ proteins in each region of interest (ROI) or Area of Illumination (AOI).

GeoMx DSP with nCounter or Next Gen Sequencer Workflow

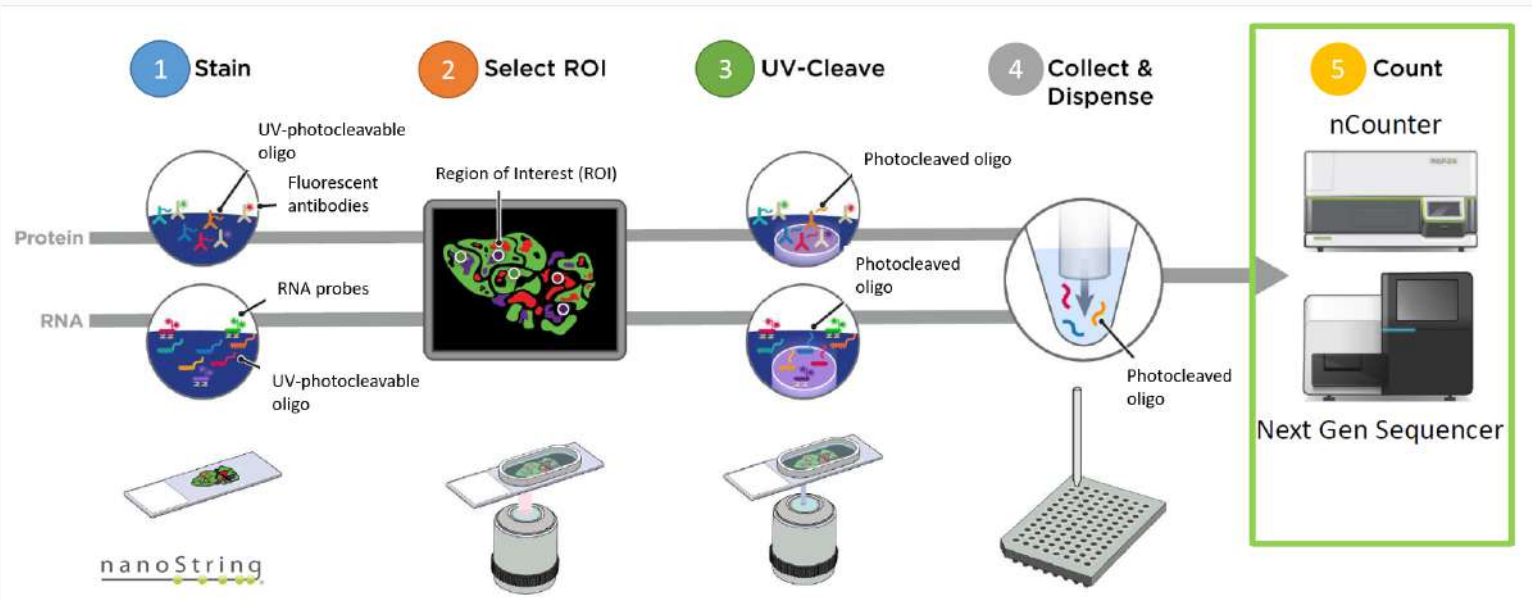

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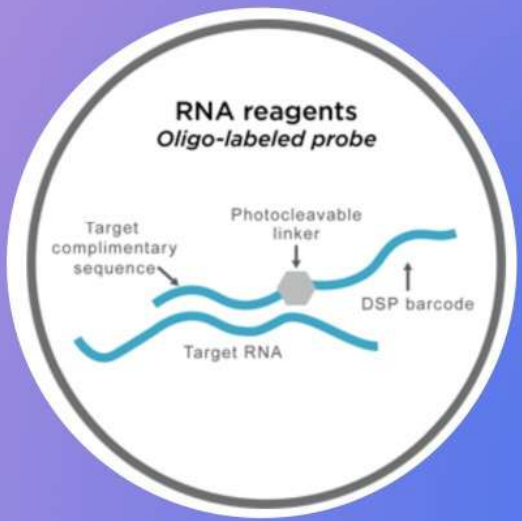
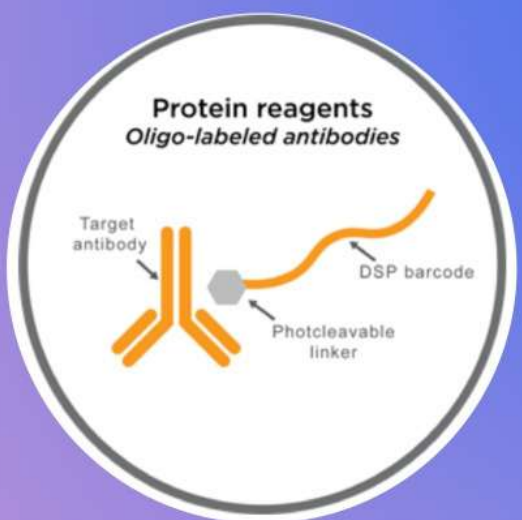
Image courtesy ; NanoString Technologies

GeoMx® RNA **Assays**



GeoMx® Human Whole Transcriptome Atlas
GeoMx® Mouse Whole Transcriptome Atlas
GeoMx® Cancer Transcriptome Atlas
GeoMx® Immune Pathways Panel
GeoMx® Custom RNA Panel

GeoMx® Protein **Assays**



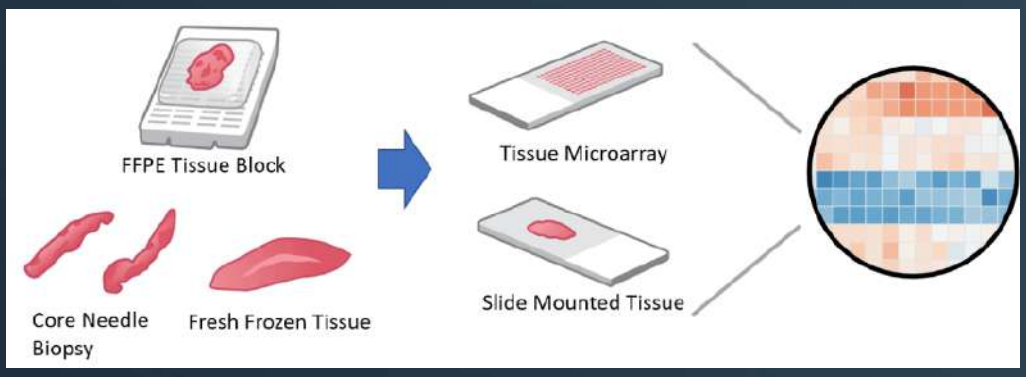
Immuno-Oncology Protein Panel
Neuroscience Protein Panel
GeoMx® Human Protein assays for NGS/ nCounter
GeoMx® Mouse Protein assays for NGS/ nCounter
GeoMx® Custom Protein Panel

KEY BENEFITS

- Highly multiplexed protein/whole transcriptome RNA detection
- Minimum sample: protein/RNA analysis from a single 5 µM FFPE section
- Morphological context: whole slide 4-color imaging to guide profiling
- Tunable resolution: select phenotypic and/or geometric regions of interest (ROI)
- Digital quantitation: up to 6 logs (base 10) dynamic range
- Data analysis: publication ready figures provided plus access to the DSP Data Analysis Suite.

Image courtesy ; NanoString Technologies

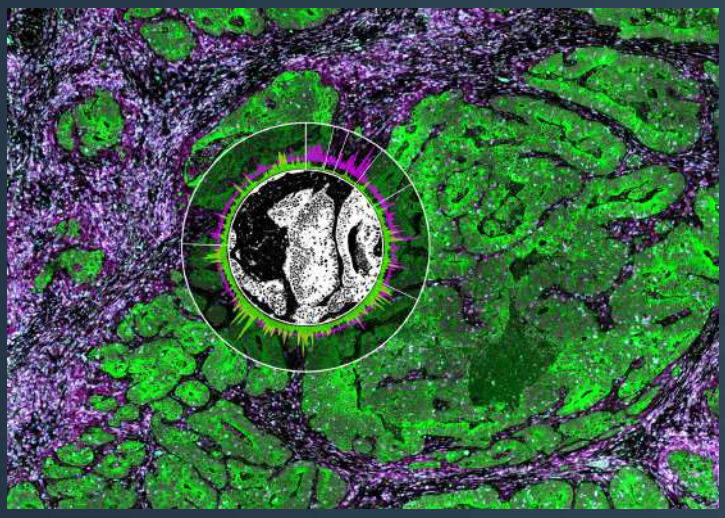
SAMPLE TYPES IN DSP



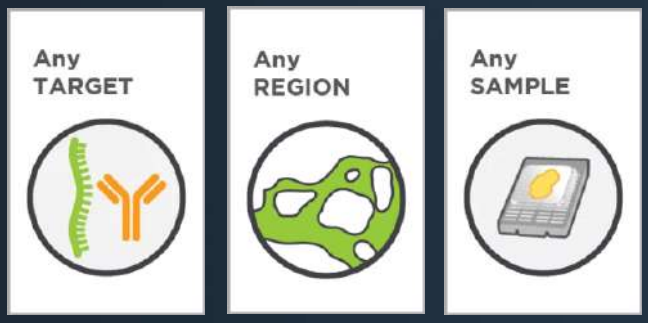
MORPHOLOGY MARKERS™

Morphology markers help reveal the structural architecture of tissue and identify regions of interest (ROI).

With GeoMx™ DSP, fluorescently labeled antibodies and/or RNA probes can be used to inform the presence of anatomical structures, cell types, and compartments, which helps with the selection and segmentation of ROIs into distinct biological compartments for further high-plex digital profiling.



BE THE FIRST TO SEE



For services and solutions please write to cues@theracues.com

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